OB1 vs. mem0 Memory Layer Division & OmniRoute Coordination

Mined from: `PLUGINS TOOLS AND MEMORY LAYER ARGUMENTS (Files 7, 8, 9, 11)`

Executive Resolution: The Dual Memory Engine

Historically debated as competing systems, OB1 (Open Brain Protocol) and mem0 serve two fundamentally different, complementary tiers in `#d.u.m.b.a.s.s.`:

Dimension	OB1 (Open Brain Protocol)	mem0
Primary Nature	Structural & Governed Access Protocol	Episodic & Adaptive Personalization
Storage Engine	PostgreSQL + `pgvector` on Node Beta	SQLite + in-memory vector cache on Node Alpha
Core Responsibility	Governs how memory is captured, classified, and doled out. Enforces time-slice proofs, session forks, and ground-truth specs.	Tracks what the user said 2 minutes ago, active session preferences, and syntax habits.
Access Interface	Model Context Protocol (MCP) Server (`knowledge.retrieve`, `knowledge.record`)	Local SDK / tool invocation hooks directly in prompt loops

OmniRoute: The Unifying Gateway

OmniRoute sits on `http://localhost:20128/v1` in front of both backends:

1. **Dynamic Dispatch:** Agents do not manually choose between OB1 and mem0. OmniRoute inspects query intent and routes to the appropriate memory provider.
2. **Local Fallback:** OmniRoute's embedded SQLite FTS5 index and typed-decay vector cache act as a zero-latency fallback when offline.
3. **Asynchronous Memory Tap:** Every completion passing through OmniRoute is tapped to update mem0 habits and send failure traces to the Reasoning Bank.